Day Ahead Residential Energy Forecasting

Comparative Modelling and Portfolio Prediction

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Forecasting Strategies

A - Weather- and Context-Aware Gradient Boosting: historical weather -> next-day temperature, sunshine and humidity forecasts -> recent consumption + household context -> HistGradientBoostingRegressor.

B - Consumption-Lag Ridge Forecasting: historical 15-minute demand + same-time day/week lags + cyclic time and weekday -> global Ridge regression.

Use of Available Information

Information	Approach A	Approach B
Consumption and calendar	7-day behaviour; week/previous-available lags; 28-day profile; calendar	Day/week lags; cyclic time/weekday
Weather and household context	Predicted weather; building, heat-pump/heating/DHW metadata	Consumption and time features
PV/EV information	Evaluated; included in cold-start model	Not used
Sparse or excluded households	Global generalisation; cold-start model	Week/day fallback; historical profile; portfolio imputation
Uncertainty	Empirical residual quantiles	Split conformal for Ridge intervals

A creates additional usable information by forecasting tomorrow's weather. Temperature affects heat-pump electricity, sunshine can influence grid import in PV homes, and recent demand captures behavioural changes; direct PV generation is not measured. Both use global models, with explicit estimates for consumers outside model fitting.

Experimental Performance

A's frozen full-year TEST (28 Feb 2023-27 Feb 2024): **portfolio WAPE 14.29%**, versus **14.47%** for persistence; household **MAE/RMSE improve 7.7%/16.1%**. Temperature MAE is **2.50°C**; nominal 80% P10-P90 portfolio interval coverage is **79.0%**. B's historical results below assess three predefined dates, rather than an equivalent full-year test.

Representative Day Ahead Tests

Energy and absolute daily error are in **MWh**; error % is signed. Each method is scored against actual demand for its own observed household set. Household coverage differs, so these results illustrate behaviour rather than establish an identical-support ranking. Missing actual demand is not treated as zero.

Date	Model	Homes	Actual	Forecast	Error %	Abs. error
2023-03-15	A	368	13.704	11.379	-16.96%	2.325
2023-03-15	B	294	11.164	9.753	-12.64%	1.411
2023-09-05	A	381	5.135	6.048	+17.79%	0.914
2023-09-05	B	374	5.047	7.154	+41.74%	2.107
2024-02-27	A	374	11.344	12.104	+6.70%	0.760
2024-02-27	B	371	11.257	11.214	-0.38%	0.043

Both methods underpredict in March and overpredict in September. B closely matches daily energy on 27 February; its -0.38% daily error does not imply the same accuracy at every 15-minute interval.

Final Day Ahead Forecast 28 February 2024

Full portfolio result	Approach A	Approach B
Households forecast	**410**	**410**
Daily point forecast	**13.015 MWh**	**12.012 MWh**
Daily uncertainty	P10 **10.470**; P50 **12.381**; P90 **14.681 MWh**	No daily portfolio bounds available

B uses 372 Ridge forecasts, 11 previous-week profiles, 23 historical profiles and 4 portfolio imputations. Both totals include all 410 consumers. This is the first genuinely unseen next-day forecast after the available record; no actual demand or forecast error is available.

Key Finding

Ridge prioritises simplicity and recurring demand patterns; gradient boosting explores adaptability through predicted weather and household context. Both forecast the entire portfolio; no ensemble is imposed.

Limitations

Historical household support and information cutoffs differ, limiting direct accuracy comparisons. A's daily uncertainty coverage is 73.4%, below nominal 80%; B's conformal intervals apply to Ridge household predictions rather than the full daily portfolio.